On Distinguishing Trees by Their Chromatic Symmetric Functions:
A Report on Martin–Morin–Wagner

Math 725 Independent Project

Abstract

This report is an exposition of, and a small extension to, the paper *On Distinguishing Trees by Their Chromatic Symmetric Functions* by J. L. Martin, M. Morin, and J. D. Wagner [9]. The central question, raised by Stanley [11], is whether the chromatic symmetric function $\mathbf{X}_T$ is a complete isomorphism invariant on the class of trees. After developing the necessary background, we present a complete proof, in our own words, of the paper's caterpillar reconstruction theorem (Theorem 5.1), which shows that caterpillars with strictly positive and pairwise distinct leaf numbers are determined up to isomorphism by their chromatic symmetric functions. We then work out two examples in detail and propose a strengthening of the paper's main open question, supported by computational evidence and a partial result.

1 Introduction

A central theme in algebraic graph theory is the search for combinatorial invariants that distinguish graphs up to isomorphism. The *chromatic polynomial* $\chi_G(k)$, which counts proper colorings of G with at most k colors, is well known but extremely weak on trees: every tree on n vertices satisfies $\chi_T(k) = k(k-1)^{n-1}$, so the chromatic polynomial cannot distinguish two non-isomorphic trees. In 1995, Stanley [11] introduced a much finer invariant.

Definition 1.1. Let $G = (V, E)$ be a finite simple graph. The *chromatic symmetric function* of G is

$$\mathbf{X}_G(x_1, x_2, \ldots) \;=\; \sum_{\kappa} \prod_{v \in V} x_{\kappa(v)},$$

where the sum runs over all proper colorings $\kappa\colon V \to \mathbb{P}$ of G by positive integers. Permuting the variables permutes the colorings, so $\mathbf{X}_G$ is a symmetric function, homogeneous of degree $|V|$.

The chromatic polynomial is recovered from $\mathbf{X}_G$ by setting $x_1 = \cdots = x_k = 1$ and $x_i = 0$ for $i > k$. Thus $\mathbf{X}_G$ refines the chromatic polynomial. Stanley showed that $\mathbf{X}_G$ is *not* a complete isomorphism invariant on graphs in general, exhibiting two non-isomorphic five-vertex graphs with the same chromatic symmetric function [11]. Whether $\mathbf{X}_T$ is a complete invariant on the restricted class of *trees* remains open and is one of the most prominent unsolved problems in algebraic combinatorics. The source paper cites Tan's computational verification through 23 vertices; later work of Heil and Ji extends this verification through 29 vertices [13, 5].

The paper of Martin, Morin, and Wagner [9] attacks this question from two complementary angles. First, they show that $\mathbf{X}_T$ is at least as strong an invariant as the *subtree polynomial* $\mathbf{S}_T(q, r)$, which is the bivariate generating function counting subtrees of T by their numbers of edges and leaves. In particular, the path and degree sequences of T can be extracted from $\mathbf{X}_T$. Second, they prove that $\mathbf{X}_T$ *does* reconstruct T when T is restricted to certain natural sub-classes, including all spiders, several families of caterpillars, and (in the unicyclic setting) all squids. The

paper's main theorem, which recovers the subtree and connector polynomials from $\mathbf{X}_T$, is the main engine behind these results: it shows that $\mathbf{X}_T$ remembers many ordinary structural statistics of a tree. The caterpillar theorem proved below is a more refined application of the same coefficient data, using the special way edge subsets sit inside a caterpillar.

The remainder of this report is organized as follows. Section 2 fixes notation and reviews symmetric-function preliminaries. Section 3 states Stanley's power-sum expansion of $\mathbf{X}_G$ and explains why it is especially well-behaved on trees. Section 4 catalogs the basic graph invariants recoverable from $\mathbf{X}_T$. Section 5 contains the main theorem of this report—a caterpillar reconstruction theorem due to Martin–Morin–Wagner—together with a complete, self-contained proof in our own words. Section 6 works out two original examples. Section 7 states a conjecture, supported by partial results, and Section 8 briefly comments on computational verification.

2 Preliminaries on symmetric functions and trees

We follow [8, 12] for symmetric functions and [1] for graphs.

Partitions. A *partition* of n is a weakly decreasing sequence $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell)$ of positive integers summing to n. We write $\lambda \vdash n$. The number $\ell = \ell(\lambda)$ is the *length* of λ. We say λ is *singleton-free* if every part is ≥ 2.

Power-sum symmetric functions. For each $k \in \mathbb{P}$ define $p_k = \sum_{i\geq 1} x_i^k$, and for a partition λ set $p_\lambda = \prod_{j=1}^{\ell(\lambda)} p_{\lambda_j}$. The set $\{p_\lambda : \lambda \vdash n\}$ is a $\mathbb{Q}$-vector-space basis for the space Λ^n of homogeneous symmetric functions of degree n.

Trees. A *tree* $T = (V, E)$ is a connected graph with $|E| = |V| - 1$; equivalently, a connected acyclic graph. A *leaf* is a vertex of degree 1, and the unique edge incident to a leaf is a *leaf edge*. We write $\delta_j(T)$ for the number of degree-j vertices of T, and $\pi_i(T)$ for the number of paths in T with exactly i edges.

Caterpillars. A *caterpillar* is a tree whose non-leaf vertices induce a nontrivial path, called the *spine*. We label the spine as $v_0, v_1, \ldots, v_s$ (so it has $s+1$ vertices and s edges), and denote by f_i the number of leaves attached to v_i, called the *i-th leaf number* of T. Note that $f_0, f_s \geq 1$ (otherwise v_0 or v_s would itself be a leaf and not lie on the spine), but f_i for $0 < i < s$ may be zero.

3 Stanley's expansion and tree-specific structure

The starting point of [9] is the following expansion of $\mathbf{X}_G$ due to Stanley [11, Thm. 2.5].

Theorem 3.1 (Stanley's expansion)**.** *For any finite graph $G = (V, E)$,*

$$\mathbf{X}_G \;=\; \sum_{A\subseteq E} (-1)^{|A|}\, p_{\mathrm{type}(A)}, \tag{1}$$

where $\mathrm{type}(A)$ *is the partition whose parts are the orders of the connected components of the spanning subgraph* (V, A)*.*

We write $c_\lambda(G)$ for the coefficient of p_λ in the power-sum expansion of $\mathbf{X}_G$, so that $\mathbf{X}_G = \sum_{\lambda \vdash n} c_\lambda(G)\, p_\lambda$.

When $G = T$ is a tree, every $A \subseteq E(T)$ is acyclic, so the spanning subgraph (V, A) has exactly $n - |A|$ components, and hence $\ell(\mathrm{type}(A)) = n - |A|$. In particular all edge subsets of a fixed type contribute the same sign $(-1)^{|A|}$, and we obtain the clean formula

$$c_\lambda(T) \;=\; (-1)^{n-\ell(\lambda)} \cdot \big|\{A \subseteq E(T) : \mathrm{type}(A) = \lambda\}\big|. \tag{2}$$

Equation (2) is the combinatorial linchpin of the entire paper, and we shall use it repeatedly.

4 Recovering classical invariants

We summarize the basic invariants of a graph (or tree) that are captured by $\mathbf{X}_G$, following [9, Prop. 2 & Cor. 5].

Proposition 4.1. *For any graph $G = (V, E)$, the chromatic symmetric function $\mathbf{X}_G$ determines:*

(i) *the number of vertices, $n = |V|$;*

(ii) *the number of edges, $|E| = -c_{(2,1,1,\ldots,1)}(G)$;*

(iii) *the number of connected components,* $\min\{\ell(\lambda) : c_\lambda(G) \neq 0\}$.

If T is a tree, then $\mathbf{X}_T$ also determines:

(iv) *the number of j-vertex subtrees, namely $\big|c_{(j,1,\ldots,1)}(T)\big|$ for each $j \geq 2$;*

(v) *the number of leaves of T, namely $|c_{(n-1,1)}(T)|$;*

(vi) *the path sequence $(\pi_i(T))_{i\geq 1}$ and the degree sequence $(\delta_j(T))_{j\geq 1}$.*

The recovery of the path and degree sequences is in fact a corollary of the much stronger result of Martin–Morin–Wagner that the *subtree polynomial* $\mathbf{S}_T(q, r) = \sum_S q^{|E(S)|}\, r^{|L(S)|}$, where the sum is over edge-nonempty subtrees S of T and $L(S)$ denotes the leaf edges of S, can be extracted from $\mathbf{X}_T$ by an explicit linear formula [9, Thm. 1]. The path sequence appears as the coefficients of $q^i r^2$ in $\mathbf{S}_T$, and the star sequence (equivalent to the degree sequence) as the coefficients of $q^k r^k$.

5 The main theorem: caterpillars with distinct leaf numbers

The principal result we shall prove is the caterpillar reconstruction theorem of [9, Thm. 11]. It illustrates beautifully how the combinatorial interpretation (2) of $c_\lambda(T)$ on trees converts a structural question—"which tree is T?"—into the combinatorial problem of identifying which edge-subsets realize which types.

Theorem 5.1 (Martin–Morin–Wagner)**.** *Let T be a caterpillar with spine $v_0, v_1, \ldots, v_s$ and leaf numbers $f_0, f_1, \ldots, f_s$. Suppose every f_i is strictly positive and that the f_i are pairwise distinct. Then T is determined up to isomorphism by its chromatic symmetric function $\mathbf{X}_T$.*

Proof. Fix the caterpillar T as in the statement, and set $n = |V(T)| = (s + 1) + \sum_{i=0}^{s} f_i$ and $L \subseteq E(T)$ the set of leaf edges of T. Write $E_{\mathrm{spine}} = \{e_1, \ldots, e_s\}$ for the spine edges, where $e_i = v_{i-1}v_i$. Throughout, partitions are equated with their multisets of parts.

We will show that the data

$$\mathcal{D}(T) \;:=\; \big\{c_\lambda(T) : \lambda \vdash n\big\}$$

suffices to recover $(f_0, f_1, \ldots, f_s)$ up to reversal. The isomorphism type of T depends only on this sequence up to reversal (reflecting the spine end-to-end produces an isomorphic caterpillar), so this is enough.

Step 1: Edge subsets with singleton-free type are exactly the supersets of L. Suppose $A \subseteq E(T)$ has type(A) singleton-free, i.e. every component of the spanning subgraph (V, A) has order ≥ 2. In particular, every leaf w of T lies in a component of size ≥ 2, which forces A to contain the unique edge incident to w. Hence $L \subseteq A$.

Conversely, if $A \supseteq L$, then for every spine vertex v_i the component of (V, A) containing v_i contains at least the $f_i \geq 1$ leaves attached to v_i together with v_i itself, so it has order $\geq f_i + 1 \geq 2$. Every leaf is also in such a component (the same one as its spine neighbor), so every component has order ≥ 2 and type(A) is singleton-free.

Consequently, edge sets A contributing to a singleton-free $c_\lambda(T)$ are precisely those of the form $A = L \cup B$ with $B \subseteq E_{\text{spine}}$.

Step 2: Identifying $\lambda :=$ type(L). Starting from the spanning subgraph (V, L), the $s+1$ nontrivial components have vertex sets $\{v_i\}$ together with the leaves attached to v_i. Adding the $|B|$ spine edges in B merges exactly $|B|$ adjacent pairs of these components, since the spine is a path and hence B is acyclic. Therefore $\ell(\text{type}(L \cup B)) = (s+1) - |B|$, and the maximum length is achieved exactly when $B = \emptyset$.

Set $\lambda :=$ type(L). Then λ has parts $f_0 + 1, f_1 + 1, \ldots, f_s + 1$ (one for each spine vertex). Step 1 and the preceding paragraph show:

$$\lambda \text{ is the } \textit{unique} \text{ singleton-free partition of } n \text{ of length } s+1 \text{ with } c_\lambda(T) \neq 0,$$

and indeed $c_\lambda(T) = (-1)^{n-(s+1)}$ since exactly one edge set ($A = L$ itself) has type λ. We can read off $s+1$ as the maximum length of a singleton-free partition appearing in $\mathbf{X}_T$ with nonzero coefficient, and then λ as the unique such partition of that length. In particular, the multiset $\{f_0 + 1, f_1 + 1, \ldots, f_s + 1\}$, hence the multiset $\{f_0, f_1, \ldots, f_s\}$, is determined by $\mathbf{X}_T$.

Note that since the f_i are pairwise distinct, all parts of λ are pairwise distinct.

Step 3: Identifying the partitions $\mu_i :=$ type$(L \cup \{e_i\})$. Now consider $|B| = 1$, say $B = \{e_i\}$. The edge $e_i = v_{i-1}v_i$ merges precisely the two components of (V, L) containing v_{i-1} and v_i, of sizes $f_{i-1} + 1$ and $f_i + 1$, into a single component of size $f_{i-1} + f_i + 2$; the other $s - 1$ components are unaffected. Hence

$$\mu_i \;:=\; \text{type}(L \cup \{e_i\}) \;=\; \big(\lambda \setminus \{f_{i-1} + 1,\ f_i + 1\}\big) \cup \{f_{i-1} + f_i + 2\},$$

where we manipulate λ as a multiset of parts.

By Step 1 and the length count from Step 2, the edge subsets with type singleton-free of length exactly s are precisely $L \cup \{e_1\}, \ldots, L \cup \{e_s\}$. Thus the relevant partitions are exactly $\mu_1, \ldots, \mu_s$, counted with multiplicity. We next explain why, under the distinctness hypothesis, each μ_i identifies the corresponding unordered adjacent pair of leaf numbers.

Since the parts of λ are pairwise distinct, the multiset difference between λ and the largest common submultiset of λ and μ_i consists of exactly the two deleted parts $f_{i-1} + 1$ and $f_i + 1$. This remains true even if the new part $f_{i-1} + f_i + 2$ happens to equal another part of λ, because we compare multisets with multiplicity. Therefore μ_i determines the unordered pair

$$\{f_{i-1} + 1,\ f_i + 1\},$$

and hence determines the unordered pair $\{f_{i-1}, f_i\}$.

If two different spine edges produced the same partition μ, the preceding paragraph would give the same unordered pair of leaf numbers for both edges. Since the f_k are pairwise distinct, that unordered pair occurs for only one edge of the path. Hence the partitions $\mu_1, \ldots, \mu_s$ are pairwise distinct, and the set $\{\mu_1, \ldots, \mu_s\}$ is determined by $\mathbf{X}_T$ as the set of singleton-free partitions μ of n with $\ell(\mu) = s$ and $c_\mu(T) \neq 0$.

Step 4: Reconstructing the leaf-number sequence. From each partition μ_i, compare it with λ as a multiset and record the two parts of λ that are deleted. Subtracting 1 from those two parts gives exactly the unordered adjacent pair $\{f_{i-1}, f_i\}$.

Form an auxiliary graph H on the vertex set $V(H) = \{f_0, f_1, \ldots, f_s\}$ (a set of $s+1$ *distinct* non-negative integers, by hypothesis) with edges $E(H) = \{\{f_{i-1}, f_i\} : 1 \leq i \leq s\}$. Steps 2 and 3 show that $V(H)$ and $E(H)$ are determined by $\mathbf{X}_T$. But H is the path f_0–f_1–$\cdots$–f_s on $s+1$ vertices, and a path on labeled vertices is determined up to reversal by its edge set. Hence the sequence $(f_0, \ldots, f_s)$ is determined by $\mathbf{X}_T$ up to reversal, which (as noted above) determines the caterpillar T up to isomorphism. □

Remark 5.2. The hypothesis that the f_i are pairwise distinct can be weakened. The argument in Step 3 only used distinctness to ensure that whenever $\mu_i = \mu_j$ one has $\{f_{i-1}, f_i\} = \{f_{j-1}, f_j\}$. If we instead assume that for each value k, the set $\{i : f_i = k\}$ forms a contiguous subpath of indices, then the argument still allows us to identify which adjacencies $\{f_{i-1}, f_i\}$ appear as multisets, and hence to reconstruct T. This relaxation is mentioned in passing in [9, §4].

6 Worked examples

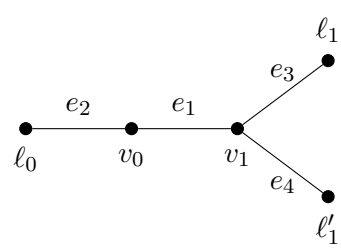
We work two examples that do not appear in [9]. The first explicitly computes the chromatic symmetric function of a small tree and verifies the identities of Proposition 4.1. The second illustrates Theorem 5.1 on a caterpillar with three spine vertices.

6.1 Example: full power-sum expansion of a five-vertex tree

Let T be the tree on $n = 5$ vertices with vertex set $\{v_0, v_1, \ell_0, \ell_1, \ell_1'\}$ and edge set $E(T) = \{e_1, e_2, e_3, e_4\}$, where

$$e_1 = v_0 v_1, \quad e_2 = v_0 \ell_0, \quad e_3 = v_1 \ell_1, \quad e_4 = v_1 \ell_1'.$$

Equivalently, T is the caterpillar with spine v_0, v_1 and leaf numbers $f_0 = 1$, $f_1 = 2$, or equivalently the spider $T_{(2,1,1)}$. A picture is shown below.



We compute $\mathbf{X}_T = \sum_{A \subseteq E(T)} (-1)^{|A|} p_{\mathrm{type}(A)}$ by enumerating all $2^4 = 16$ edge subsets $A \subseteq E(T)$ and tabulating their types.

$\lvert A\rvert$	subsets	component sizes	type
0	$\varnothing$	five singletons	(1^5)
1	each of $\{e_1\},\{e_2\},\{e_3\},\{e_4\}$	one edge plus 3 isolated	$(2,1,1,1)$
2	$\{e_1,e_2\},\{e_1,e_3\},\{e_1,e_4\},\{e_3,e_4\}$	one P_3 plus 2 isolated	$(3,1,1)$
2	$\{e_2,e_3\},\{e_2,e_4\}$	two edges plus 1 isolated	$(2,2,1)$
3	$\{e_1,e_2,e_3\},\{e_1,e_2,e_4\},\{e_1,e_3,e_4\}$	one P_4/star plus 1	$(4,1)$
3	$\{e_2,e_3,e_4\}$	disjoint P_2 and P_3	$(3,2)$
4	$E(T)$	all of T	(5)

Tallying the signed contributions to each $c_\lambda(T)$ gives

$$\begin{aligned}\mathbf{X}_T \;=\; & p_{(1,1,1,1,1)} \;-\; 4\,p_{(2,1,1,1)} \;+\; 4\,p_{(3,1,1)} \;+\; 2\,p_{(2,2,1)} \\ & -\; 3\,p_{(4,1)} \;-\; p_{(3,2)} \;+\; p_{(5)}.\end{aligned} \tag{3}$$

We now verify each part of Proposition 4.1:

(i) Every monomial of every p_λ in (3) has total degree $5 = |V(T)|$.

(ii) $-c_{(2,1,1,1)}(T) = -(-4) = 4 = |E(T)|$.

(iii) The minimum length of a partition with non-zero coefficient is $\ell((5)) = 1$, matching the 1 component of T.

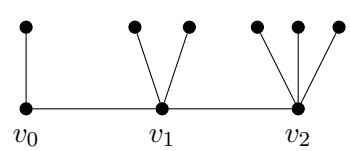
(iv) For $j = 3$ we have $|c_{(3,1,1)}(T)| = 4$, and indeed T has 4 subtrees on 3 vertices: the three paths $\ell_0 v_0 v_1$, $v_0 v_1 \ell_1$, $v_0 v_1 \ell_1'$, and the two-edge path $\ell_1 v_1 \ell_1'$. Similarly $|c_{(2,2,1)}|$ does *not* count subtrees, only single-component contributions from edge sets of type $(2,2,1)$.

(v) $|c_{(4,1)}(T)| = 3$, which equals the number of leaves of T (namely ℓ_0, ℓ_1, ℓ_1').

The path sequence and degree sequence of T are also visible: $\delta_1(T) = 3$ (leaves), $\delta_2(T) = 1$ (vertex v_0), $\delta_3(T) = 1$ (vertex v_1); the path counts are $\pi_1 = 4$, $\pi_2 = 4$, $\pi_3 = 2$, $\pi_4 = 0$. These can be read off from the corresponding coefficients in the subtree polynomial $\mathbf{S}_T(q,r)$, which Theorem 1 of [9] extracts from (3) by the linear formula given there.

6.2 Example: applying Theorem 5.1

Let T be the caterpillar with spine v_0, v_1, v_2 ($s = 2$) and leaf numbers $f_0 = 1$, $f_1 = 2$, $f_2 = 3$. This caterpillar has $n = 3 + 1 + 2 + 3 = 9$ vertices and 8 edges, and its leaf numbers satisfy the hypotheses of Theorem 5.1: all are positive and pairwise distinct.



Rather than compute the full chromatic symmetric function (which has $2^8 = 256$ edge subsets to enumerate), we extract precisely the coefficients required by the proof of Theorem 5.1.

Computing $\lambda = \text{type}(L)$. The set L of leaf edges has cardinality $f_0 + f_1 + f_2 = 6$. Removing all spine edges from T and keeping only L leaves three components, of orders $f_0 + 1 = 2$, $f_1 + 1 = 3$, $f_2 + 1 = 4$. Hence

$$\lambda \;=\; (4, 3, 2), \qquad c_\lambda(T) = (-1)^{9-3} = +1.$$

We verify that λ is the unique singleton-free partition of 9 of length 3 with non-zero coefficient. The other singleton-free partitions of 9 of length 3 are $(5, 2, 2)$ and $(3, 3, 3)$. By Step 1 of the proof, every A with singleton-free type contains L (forcing the components to be unions of the three blocks $\{v_0, \text{leaf}\}$, $\{v_1, \text{leaves}\}$, $\{v_2, \text{leaves}\}$ of orders $2, 3, 4$). No subset of these blocks can be merged to produce component sizes $\{5, 2, 2\}$ or $\{3, 3, 3\}$, so $c_{(5,2,2)} = c_{(3,3,3)} = 0$.

Computing the μ_i. The two spine edges are $e_1 = v_0v_1$ and $e_2 = v_1v_2$.

- $L \cup \{e_1\}$ merges the components of orders 2 and 3, yielding $\mu_1 = (5, 4)$.
- $L \cup \{e_2\}$ merges the components of orders 3 and 4, yielding $\mu_2 = (7, 2)$.

Both are singleton-free partitions of length 2, and indeed $\mu_1 \neq \mu_2$. Each contributes $(-1)^{9-2} = -1$ to its coefficient, and (as in Step 1) no other edge subset gives a singleton-free partition of length 2, so $c_{(5,4)}(T) = c_{(7,2)}(T) = -1$.

Reconstruction of T from $\mathbf{X}_T$. Suppose we are handed only the data $\mathbf{X}_T$. We extract:

(1) By Proposition 4.1(i)–(iii), $n = 9$, $|E(T)| = 8$, and the graph has one connected component. Hence it is connected with $n-1$ edges, so it is a tree on 9 vertices. By Proposition 4.1(v) and Corollary 5 of [9], $\delta_1(T) = 6$ leaves and the diameter is 4, from which one may verify (using Proposition 10 of [9]) that T is a caterpillar.

(2) The maximum length of a singleton-free partition with $c_\lambda(T) \neq 0$ is $s + 1 = 3$, so the spine has 3 vertices.

(3) The unique such partition is $\lambda = (4, 3, 2)$, so the multiset of leaf numbers is $\{4-1, 3-1, 2-1\} = \{3, 2, 1\}$.

(4) The singleton-free partitions of 9 of length 2 with $c_\lambda(T) \neq 0$ are $(5, 4)$ and $(7, 2)$, so $\{\mu_1, \mu_2\} = \{(5, 4), (7, 2)\}$.

(5) Comparing $\mu_1 = (5, 4)$ to $\lambda = (4, 3, 2)$, the multiset of parts of λ absent from μ_1 is $\{3, 2\}$. Hence the endpoints of one spine edge have leaf numbers $\{3 - 1, 2 - 1\} = \{2, 1\}$.

(6) Comparing $\mu_2 = (7, 2)$ to $\lambda = (4, 3, 2)$, the missing parts are $\{4, 3\}$, so the endpoints of the other spine edge have leaf numbers $\{3, 2\}$.

(7) The auxiliary graph H has vertex set $\{1, 2, 3\}$ and edges $\{1, 2\}$ and $\{2, 3\}$, which is the path 1–2–3. Reading off the path gives the leaf-number sequence $(1, 2, 3)$, up to reversal.

This recovers T up to isomorphism, as predicted by Theorem 5.1.

7 A conjecture and partial progress

The source paper proves reconstructibility for several special families of caterpillars, including the distinct positive leaf number case treated in Theorem 5.1. Later work proves stronger caterpillar results: proper caterpillars are distinguished by their chromatic symmetric functions [6], and Loebl and Sereni prove Stanley's isomorphism conjecture for caterpillars more generally [7]. Thus the broad statement "every caterpillar is determined by $\mathbf{X}_T$" is best regarded here as historical motivation rather than as an open conjecture.

The proof of Theorem 5.1, however, suggests a sharper question about how much of $\mathbf{X}_T$ is actually needed. Let the *singleton-free coefficient data* of a tree T mean the set of coefficients $c_\lambda(T)$ for partitions λ with no part equal to 1.

Conjecture 7.1 (Singleton-free caterpillar reconstruction)**.** *Let T be a caterpillar whose leaf numbers $f_0, \ldots, f_s$ are all positive. Then the singleton-free coefficient data of $\mathbf{X}_T$ determines the linear order of the multiset $\{f_0, \ldots, f_s\}$ along the spine, up to reversal. Equivalently, this data determines T up to isomorphism among positive caterpillars.*

Evidence for Conjecture 7.1.

1. Theorem 5.1 proves the conjecture when the leaf numbers are pairwise distinct. In that case, the length $s+1$ singleton-free partition $\lambda = \mathrm{type}(L)$ recovers the vertex set of leaf numbers, and the length s singleton-free partitions $\mu_i = \mathrm{type}(L \cup \{e_i\})$ recover the edge set of the path of leaf numbers.

2. Remark 5.2 gives additional positive cases when equal leaf numbers occur in contiguous blocks.

3. The accompanying SageMath code performs a brute-force search over positive caterpillars up to 12 vertices and finds no collision in the singleton-free coefficient data among non-isomorphic examples.

4. The known reconstructibility of caterpillars from the full chromatic symmetric function [6, 7] supports the idea that the essential information may already be concentrated in the singleton-free part of the power-sum expansion.

A partial result. The following proposition proves the first stage of the conjecture: the singleton-free coefficient data determines the multiset of leaf numbers, even if it does not yet determine their order.

Proposition 7.2. *Let T be a caterpillar with spine $v_0, \ldots, v_s$ and leaf numbers $f_0, \ldots, f_s$, and assume every $f_i \geq 1$. Then the multiset $\{f_0, f_1, \ldots, f_s\}$ is determined by the singleton-free coefficient data of $\mathbf{X}_T$.*

Proof. Let L be the set of leaf edges of T. As in Step 1 of the proof of Theorem 5.1, an edge set $A \subseteq E(T)$ has singleton-free type if and only if $L \subseteq A$. Therefore the maximum possible length of a singleton-free type is achieved when $A = L$ and is equal to $s+1$. Moreover, $A = L$ is the unique edge set giving singleton-free type of this maximum length.

Thus the singleton-free coefficient data identifies the unique singleton-free partition λ of maximum length with nonzero coefficient. This partition is

$$\lambda = \mathrm{type}(L) = (f_0 + 1, f_1 + 1, \ldots, f_s + 1)$$

as a multiset of parts. Subtracting 1 from each part of λ recovers the multiset $\{f_0, \ldots, f_s\}$. $\square$

Proposition 7.2 reduces Conjecture 7.1 to the problem of recovering the linear order, up to reversal, of a known multiset of positive leaf numbers along the spine. The proof of Theorem 5.1 solves this when the leaf numbers are distinct by using the partitions μ_i. When repeated leaf numbers occur, several edge sets may share the same type, so one has to control multiplicities in the coefficients $c_\mu(T)$. This is the main obstruction to extending the same elementary proof to all positive caterpillars.

8 Computational aspects

A natural complement to the proof and examples above is a computational verification of Theorem 5.1 and an exploration of Conjecture 7.1. The accompanying SageMath file `csf_trees.sage` implements the following:

- a routine that computes the power-sum coefficient dictionary $\{\lambda \mapsto c_\lambda(T)\}$ by enumerating all edge subsets $A \subseteq E(T)$ and applying Stanley's expansion;
- routines that compute the connector polynomial $\mathbf{K}_T(x, y)$ and the subtree polynomial $\mathbf{S}_T(q, r)$ directly from the tree;
- a verification routine for the formulas of Theorem 1 of Martin–Morin–Wagner, comparing the directly computed connector and subtree polynomials with the values obtained from the coefficients $c_\lambda(T)$;
- a reconstruction routine for positive caterpillars with distinct leaf numbers, following the four-step proof of Theorem 5.1;
- a brute-force search over positive caterpillars up to 12 vertices that compares singleton-free coefficient data and reports whether any non-isomorphic examples collide.

The code takes SageMath graph objects as input. Its output consists of coefficient dictionaries, polynomial dictionaries, reconstructed leaf number sequences, and collision reports. The README supplied with the code explains how to run the script and includes sample output for the examples discussed in this report.

References

[1] B. Bollobás, *Modern Graph Theory*, Graduate Texts in Mathematics 184, Springer-Verlag, New York, 1998.

[2] D. Eisenstat and G. Gordon, *Non-isomorphic caterpillars with identical subtree data*, Discrete Math. **306** (2006), 827–830.

[3] J. Fougere, *On symmetric chromatic polynomials of trees*, undergraduate thesis, Dartmouth College, 2003.

[4] G. Gordon and E. McDonnell, *Trees with the same degree sequence and path number*, Discrete Math. **147** (1995), 297–300.

[5] S. Heil and C. Ji, *On an algorithm for comparing the chromatic symmetric functions of trees*, Australas. J. Combin. **75** (2019), 210–222.

[6] J. Aliste-Prieto and J. Zamora, *Proper caterpillars are distinguished by their chromatic symmetric function*, Discrete Math. **315–316** (2014), 158–164.

[7] M. Loebl and J.-S. Sereni, *Isomorphism of weighted trees and Stanley's isomorphism conjecture for caterpillars*, Ann. Inst. Henri Poincaré D **6** (2019), 357–384.

[8] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford University Press, New York, 1995.

[9] J. L. Martin, M. Morin, and J. D. Wagner, *On distinguishing trees by their chromatic symmetric functions*, J. Combin. Theory Ser. A **115** (2008), 237–253.

[10] M. Morin, *Caterpillars, Ribbons, and the Chromatic Symmetric Function*, M.S. thesis, University of British Columbia, 2005.

[11] R. P. Stanley, *A symmetric function generalization of the chromatic polynomial of a graph*, Adv. Math. **111** (1995), 166–194.

[12] R. P. Stanley, *Enumerative Combinatorics, Vol. 2*, Cambridge University Press, Cambridge, 1999.

[13] L.-Y. Tan, *Computational verification that the chromatic symmetric function determines up to isomorphism trees of order at most* 23, available at http://www.cs.wustl.edu/~lt1/csf.html, retrieved 2007.